Skills Summary

Using Dilations to Reason About Similarity

Use this reference while completing your worksheet or when studying.

Skill 1 — Coordinates of a dilated point

Rules:

Centre at the origin: $(x, y) \longmapsto (kx, ky)$.

Centre at (h, m): $(x, y) \longmapsto \big(h + k(x-h),\ m + k(y-m)\big)$ — measure from the centre, multiply by k, step back out.

$k > 1$ enlarges, $0 < k < 1$ reduces, and $k = 1$ leaves the figure alone.

Example: A dilation about the origin with $k = 3$ maps $P(4, -2)$ to P'. Find the x-coordinate of P'.
Step 1. The centre is the origin, so each coordinate is just multiplied by k; take the x-coordinate on its own.
Step 2. $3 \times 4 = \mathbf{12}$ (the y-coordinate is handled the same way).

Try it: A dilation about the origin with $k = 0.5$ maps $Q(5, -4)$ to Q'. Find the y-coordinate of Q'.

check: **−2**

Skill 2 — Scale factor from two lengths

Rule:

$$k = \frac{\text{image length}}{\text{original length}}$$

Image over original, in that order: k answers "how many times as long is the image?" Lengths on a grid come from the distance formula.

Example: A model edge is 12 cm long. On the enlarged copy the matching edge is 18 cm. Find k.
Step 1. The two edges correspond, so divide the image length by the original length.
Step 2. $18 \div 12 = \mathbf{1.5}$, so every length on the copy is 1.5 times the model's.

Try it: An 8 cm edge becomes 20 cm on the enlarged copy. Find k.

check: **2.5**

Skill 3 — Slope under a dilation

Rule:

A dilation maps a segment to a **parallel** segment, so the image has the *same slope*. Equal slopes are the coordinate evidence that the angles did not change — which is why the figure and its image are similar, not just the same size.

Example: Find the slope of the segment from $(1, 2)$ to $(3, 6)$. Its image under $k = 4$ runs from $(4, 8)$ to $(12, 24)$.
Step 1. Slope is rise over run, and the claim to test is that the image gives the same number.
Step 2. $(6-2)/(3-1) = 4/2 = \mathbf{2}$, and the image gives $(24-8)/(12-4) = 16/8 = 2$ as well.

Try it: Find the slope of the segment from $(2, 1)$ to $(8, 4)$.

check: **0.5**

Skill 4 — Similar figures: sides and area

Rules:

In similar figures every pair of matching sides has the *same* ratio k, so a missing side is $k\times$ (matching side).

Lengths scale by k, but area scales by k^2: both dimensions grew, so the area grew twice.

Example: A triangle has sides 5 cm and 7 cm. A similar triangle has 15 cm matching the 5 cm side. Find the side matching 7 cm.
Step 1. Matching sides share one ratio, so find k from the pair that is fully known, then apply it to the other side.
Step 2. $k = 15 \div 5 = 3$, so the missing side is $3 \times 7 = \mathbf{21}$ cm.

Try it: A triangle has sides 4 cm and 9 cm. A similar triangle has 10 cm matching the 4 cm side. Find the side matching 9 cm (in cm).

check: **22.5**

Example: A figure has area 15 square units. It is dilated with $k = 2$. Find the area of the image.
Step 1. Area is a product of two lengths and each one is multiplied by k, so multiply the area by k^2, not by k.
Step 2. $15 \times 2^2 = 15 \times 4 = \mathbf{60}$ square units.

Try it: A figure has area 24 square units and is dilated with $k = 3$. Find the area of the image (in square units).

check: **216**